Physics of Semiconductor Devices

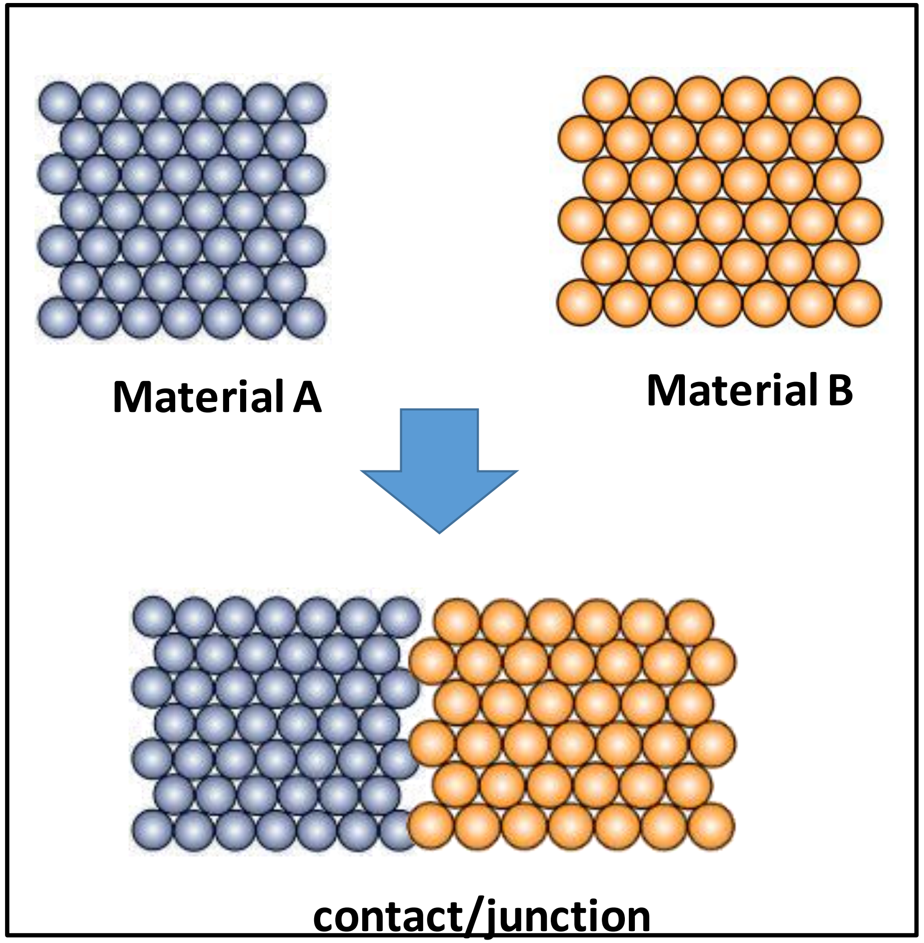
ECE210: PSD

INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

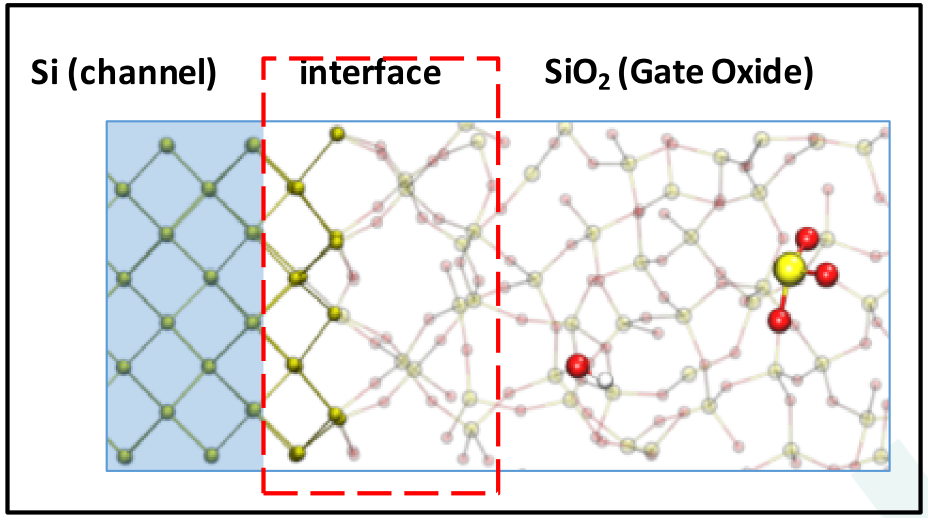
Dr. Ram Krishna Ghosh

(week 08)

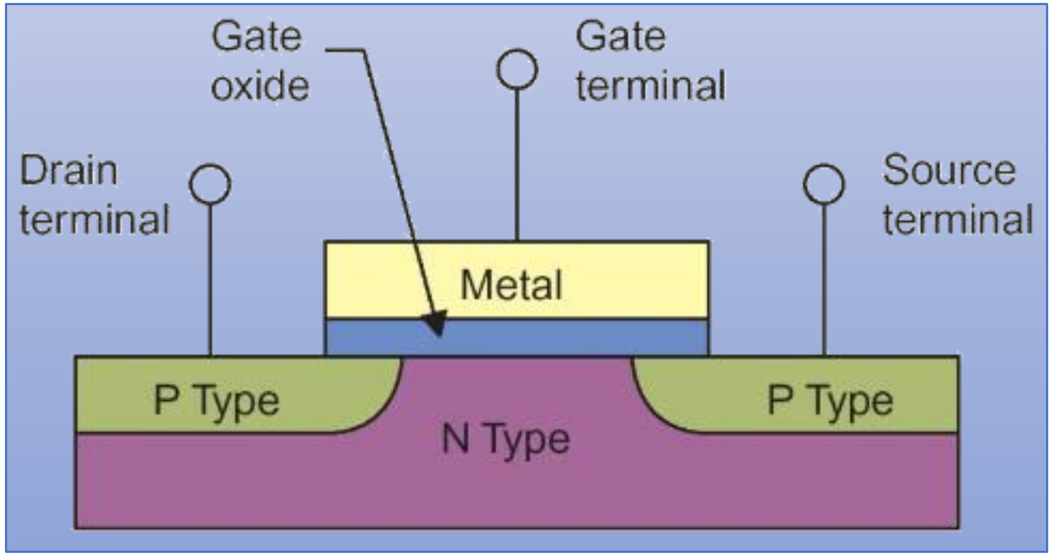
Contact/junctions



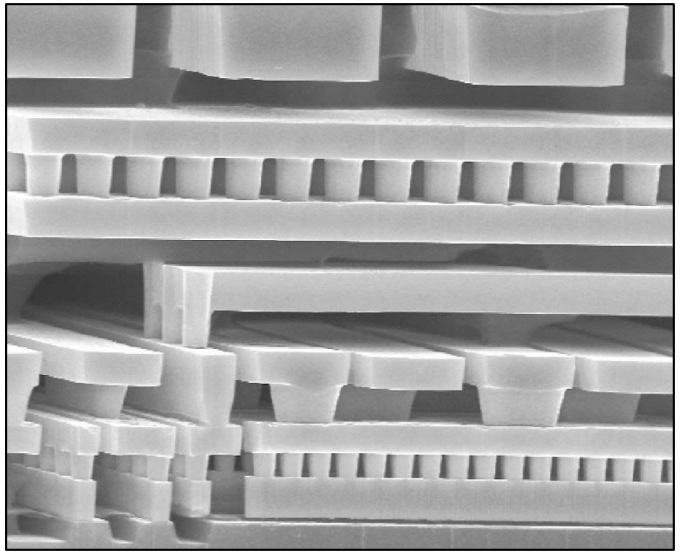
Semiconductor-oxide junction



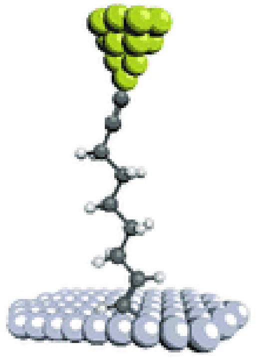
Contacts/junctions



Semiconductor devices (e.g. MOSFETs)



Multilevel Metallization (e.g. vias)

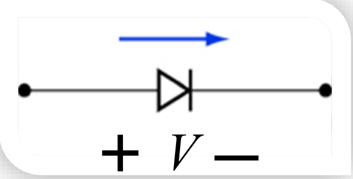


Molecular junction

Any type of electronics device has different junctions:

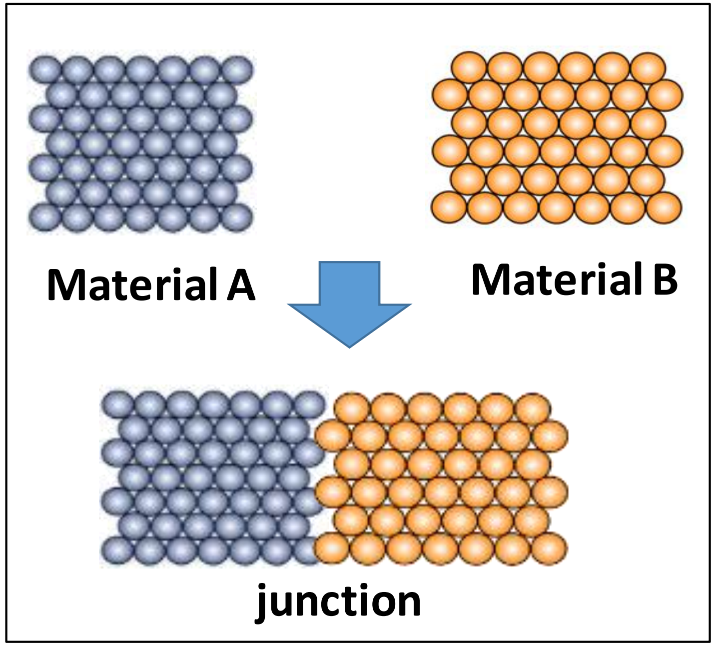
<u>such as:</u>
Metal-metal / metal semiconductor / metal semiconductor / semiconductor-semiconductor / metal –oxide –semiconductor/ metal molecule junction, etc.

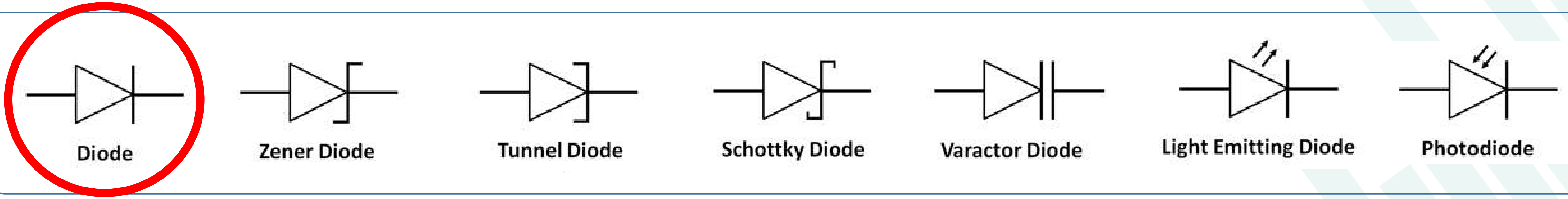
Diodes: two-terminal devices; (that conducts current primarily in one direction)



Junctions: In semiconductors, made by junctions between two different materials

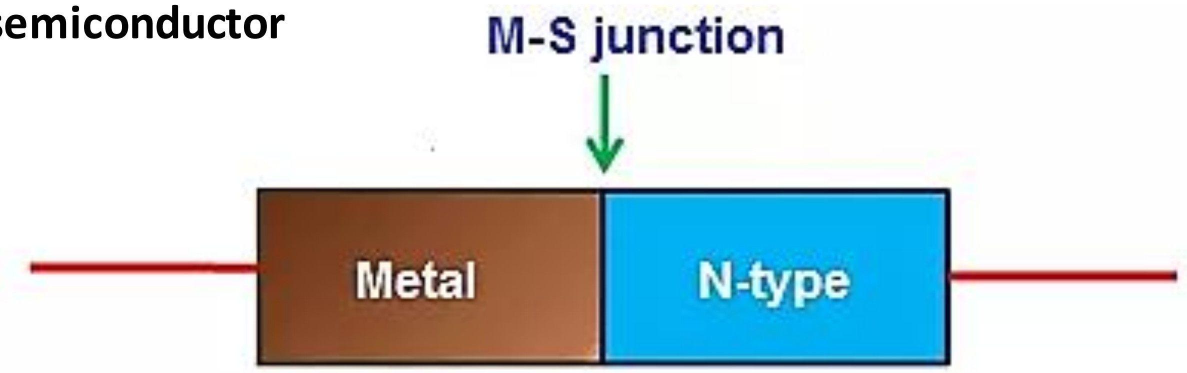
- Homojunctions: junctions between two differently doped regions of the same semiconductor material
- Heterojunctions: junctions between two different types of materials (usually means semiconductor materials)
- Metal-semiconductor junctions
  - But not all metal-semiconductor junctions are diodes



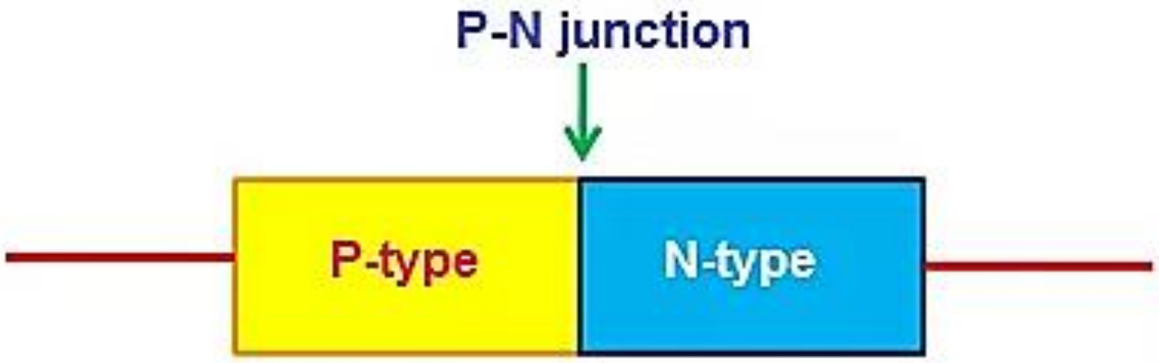


Junctions in semiconductor devices

metal semiconductor



semiconductor semiconductor

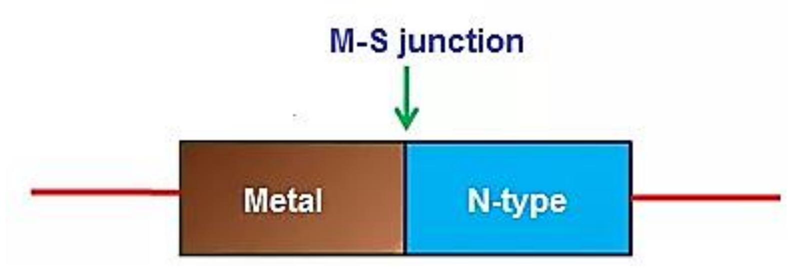


A p-n junction is not a device

A p-n junction diode is a device

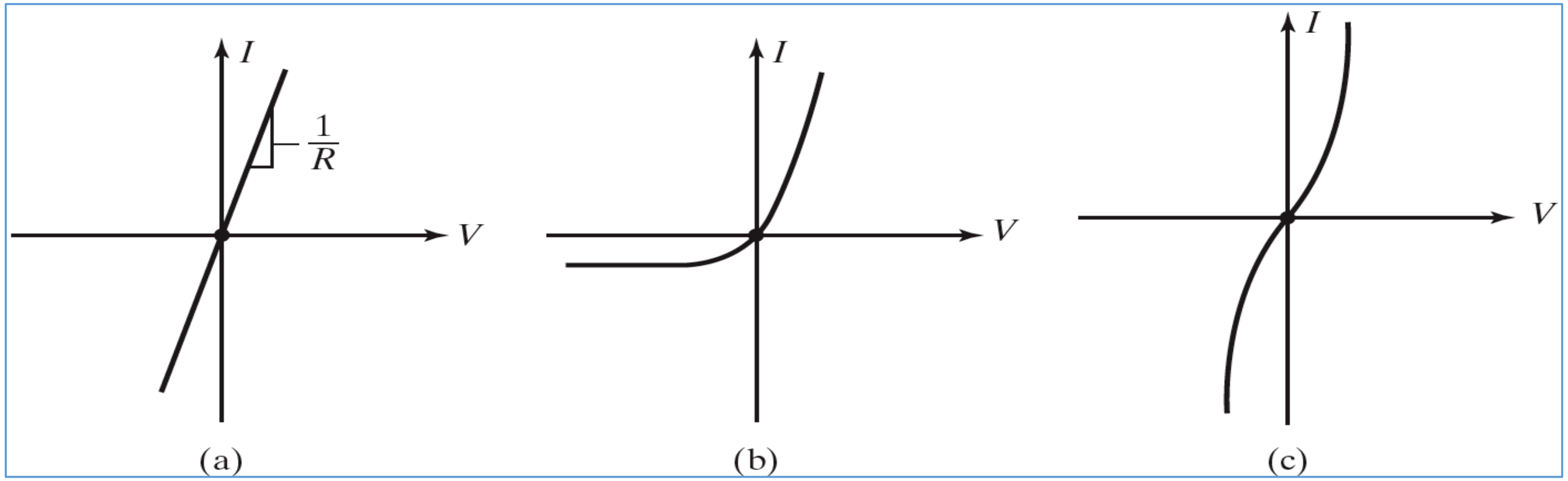
<u>Difference:</u> a p-n junction also has two Metal-Semiconductor (MS) junction

Metal-Semiconductors Contacts



Metal-semiconductor contacts are an obvious component of any semiconductor device

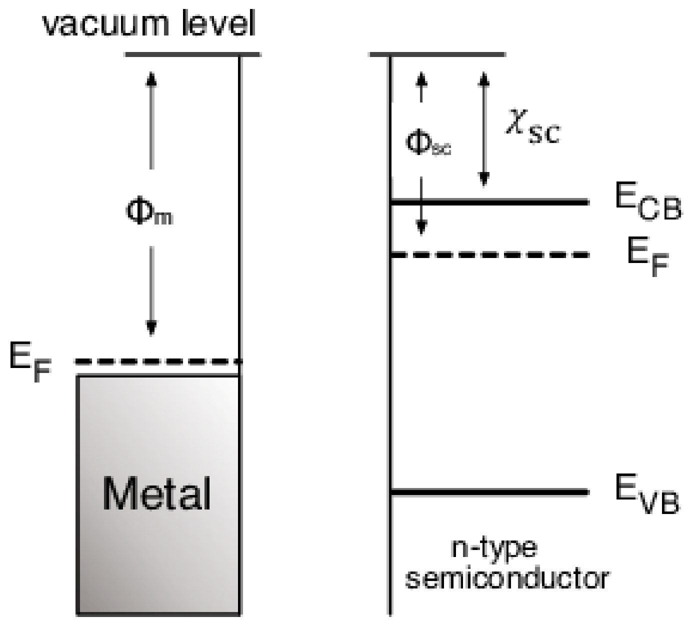
At the same time, such contacts cannot be assumed to have a resistance as low as that of two connected metals.



(a) Ideal Ohmic Contact (Gold to p-type silicon) (b) Schottky Contact (Aluminum to n-type silicon) (c) Nonlinear "Ohmic" Contact (Aluminum to n^+ silicon)	Ohmic → Non-rectifying contact Schottky → Rectifying Contact

Metal-Semiconductors Contacts

The principle of forming different types of the metal-semiconductor contact is the mismatch of the Fermi energy between metal and semiconductor material, which is due to the difference in work functions.



Primarily, the electron affinity and the metal work function Φ_M are invariant fundamental properties that are remain unaffected by the contacting process.

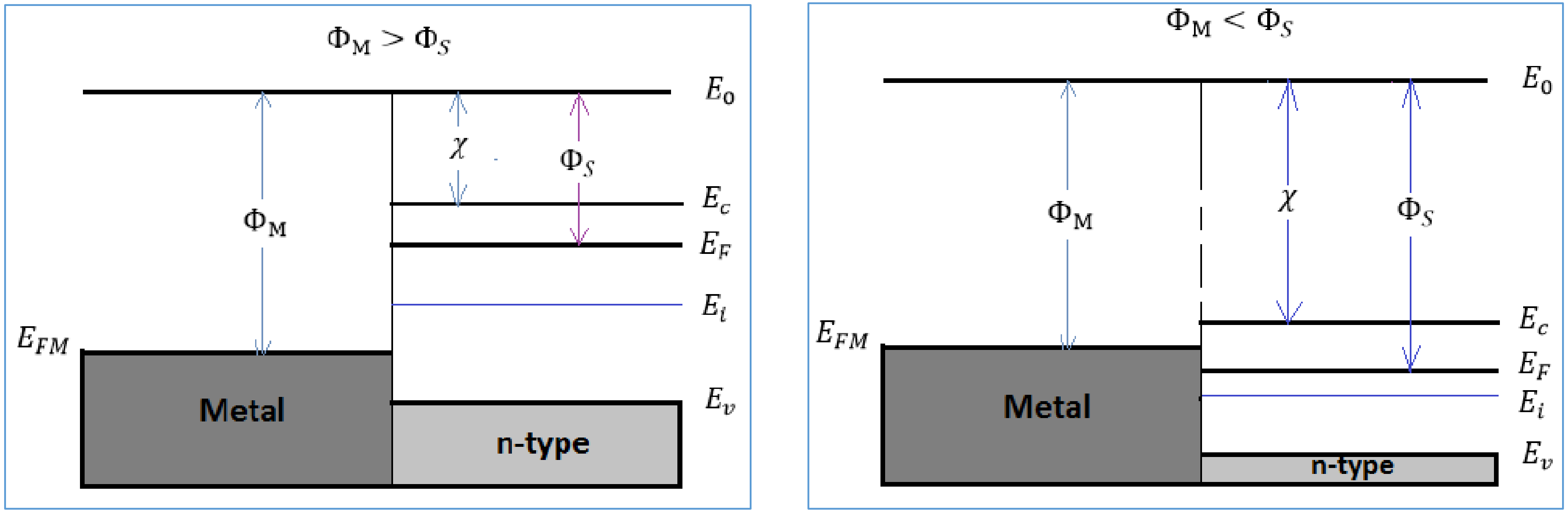
For the ideal MS contact, several assumptions are made:

1. The metal and the semiconductor are contacted intimately, which means that there is no oxide or charge layers between the contact on the atomic scale.

2. No intermixing and inter diffusion between the metal and the semiconductor.

3. There are no impurities at the MS interface.

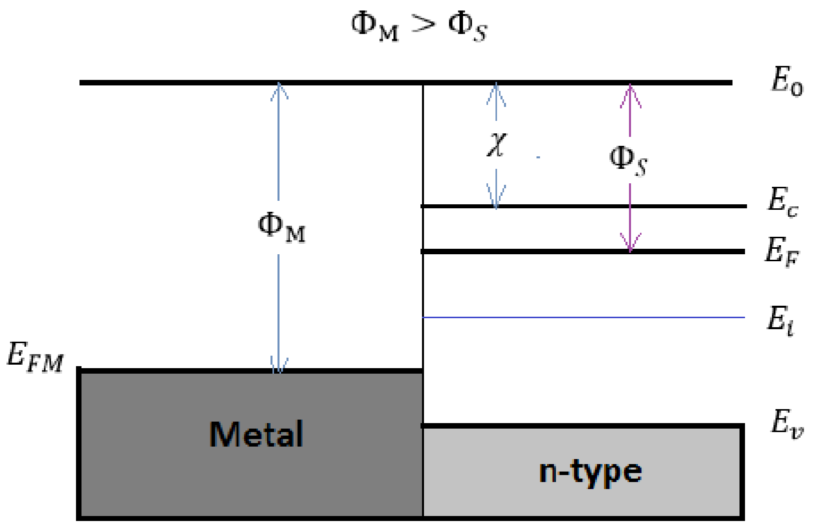
Metal-Semiconductors Contacts



Energy band diagram for ideal MS contacts at an instant after contact for: $\Phi_M > \Phi_S$ and $\Phi_M < \Phi_S$

The situations here are not in equilibrium condition, since the Fermi energy in the metal is not aligned with the Fermi energy in semiconductor ($E_{FS} \neq E_{FM}$). Therefore, electrons will keep transferring between the semiconductor and the metal until the E_F is aligned.

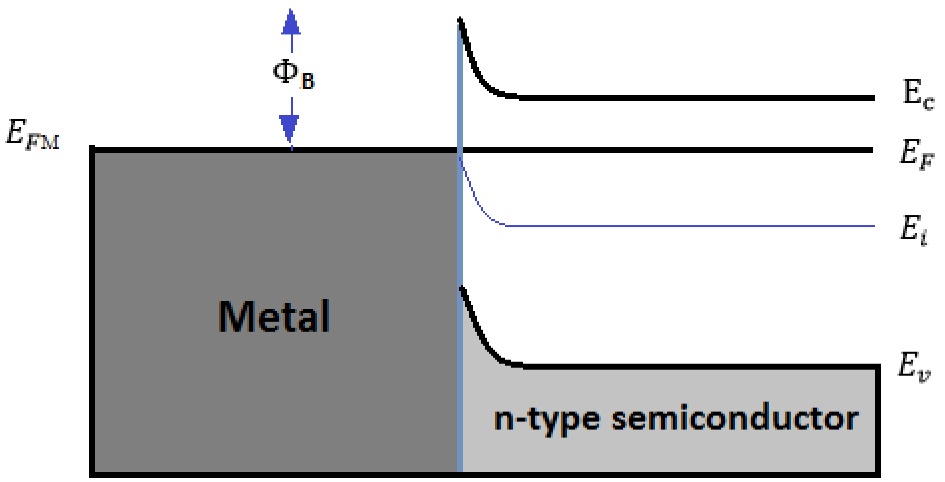
Schottky Barrier Contact Under Equilibrium Condition (applied bias V=0)



energy band diagram
immediately after the contact

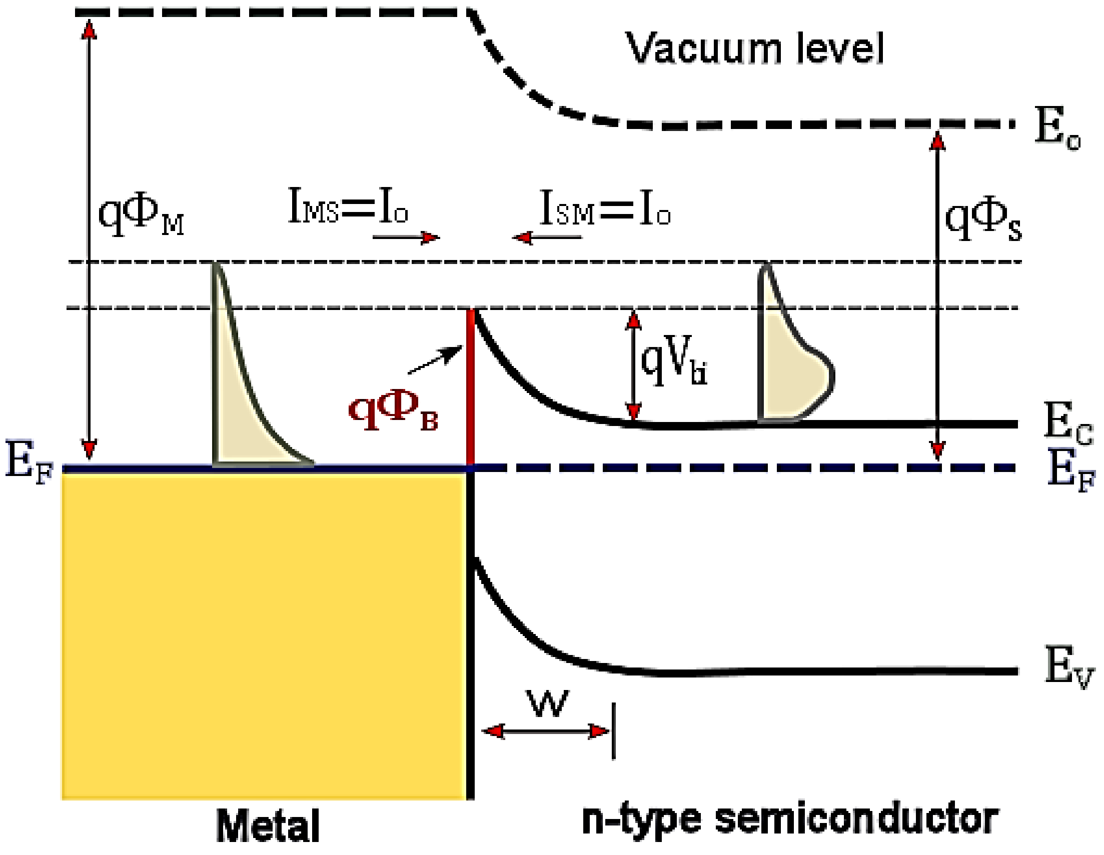
Consider the case of $\Phi_M > \Phi_S$, electrons will transfer from the semiconductor to the metal due to their greater energy until the equilibrium condition is established.

Under Equilibrium Condition



The net loss of electrons creates a negative charge in the metal and a positive charge in the semiconductor, which results a depletion region and a growing barrier at the semiconductor surface

Schottky Barrier Contact Under Equilibrium Condition (applied bias V=0)

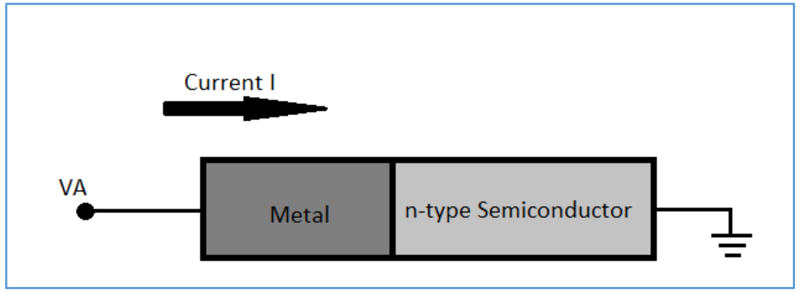


for n type Semiconductor

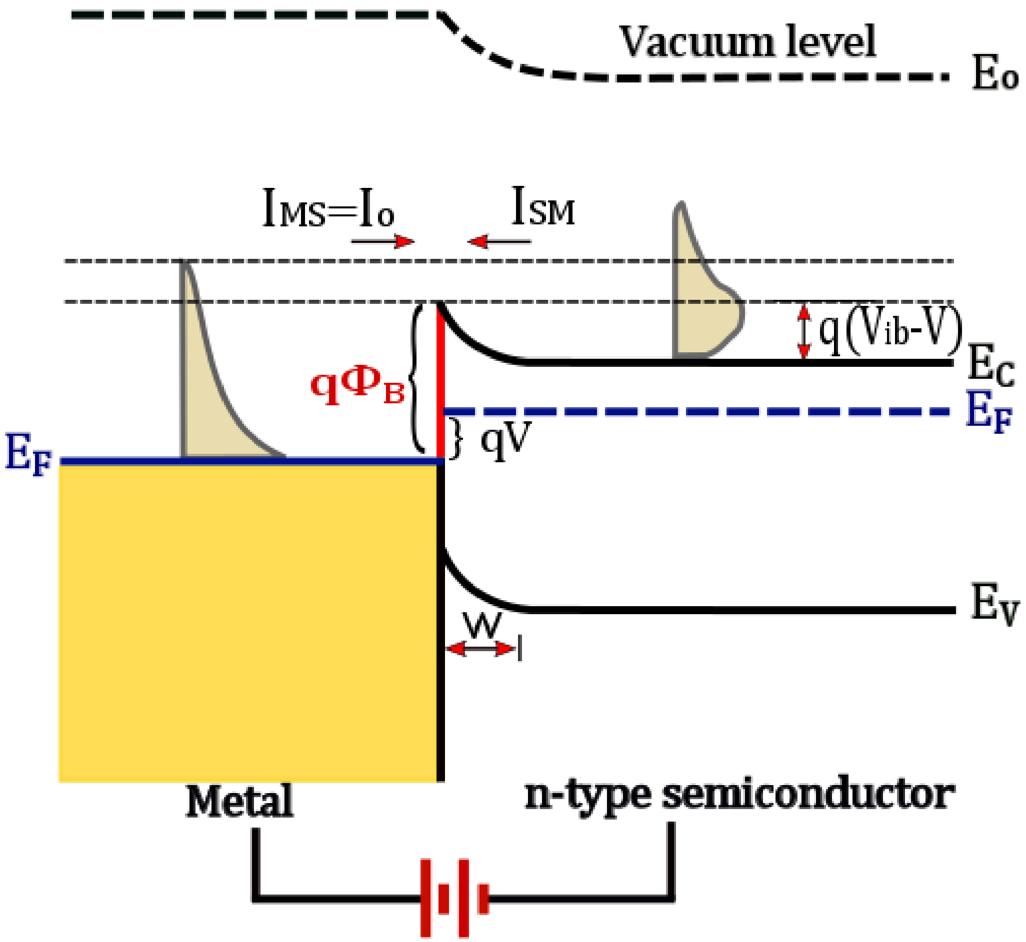
$$\Phi_B = \Phi_M - \chi$$

$$qV_{bi} = q(\Phi_M - \Phi_S)$$

Schottky Barrier Contact Under bias V

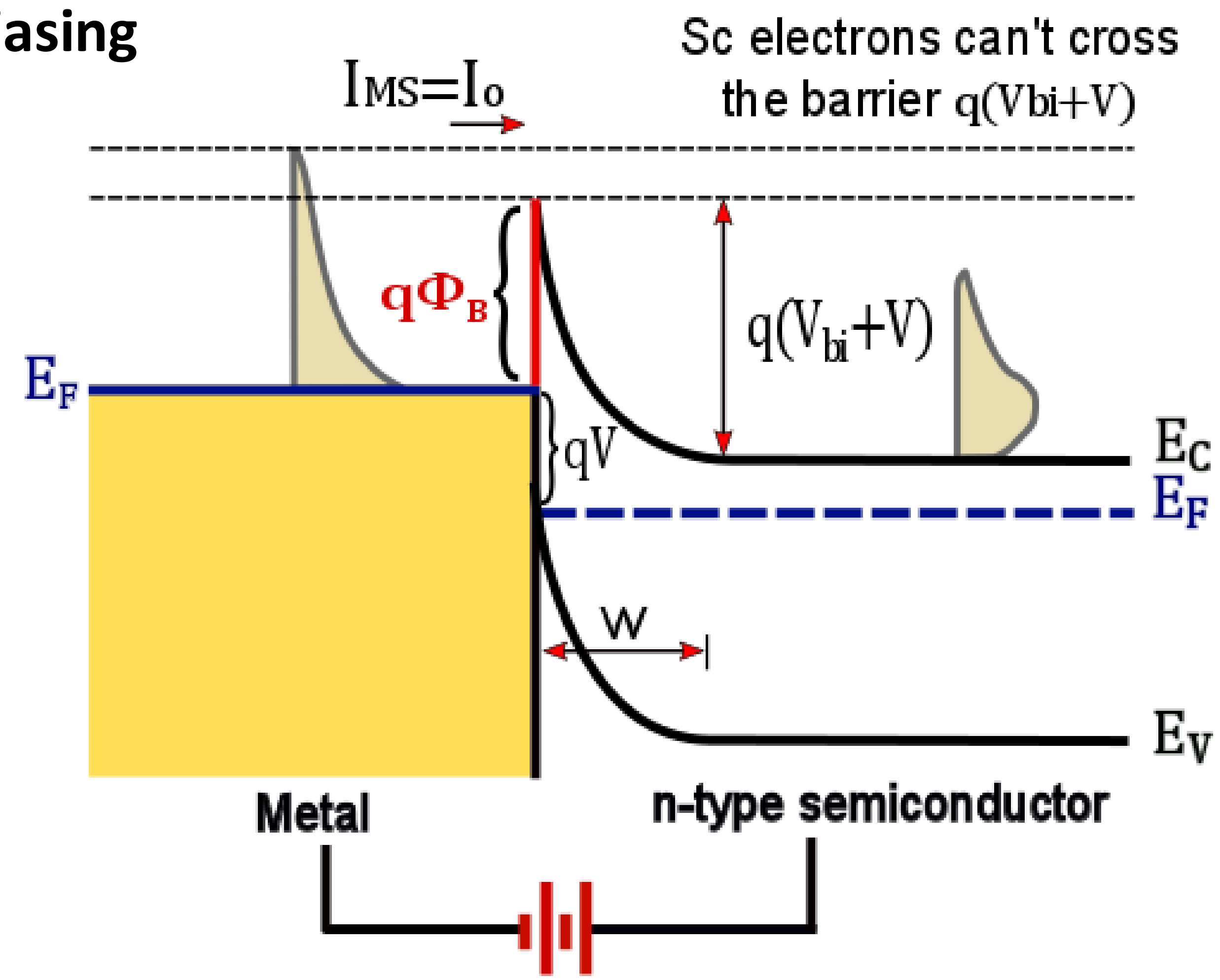


Forward Biasing

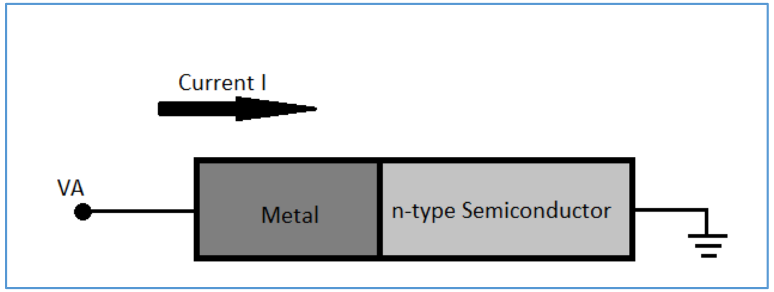


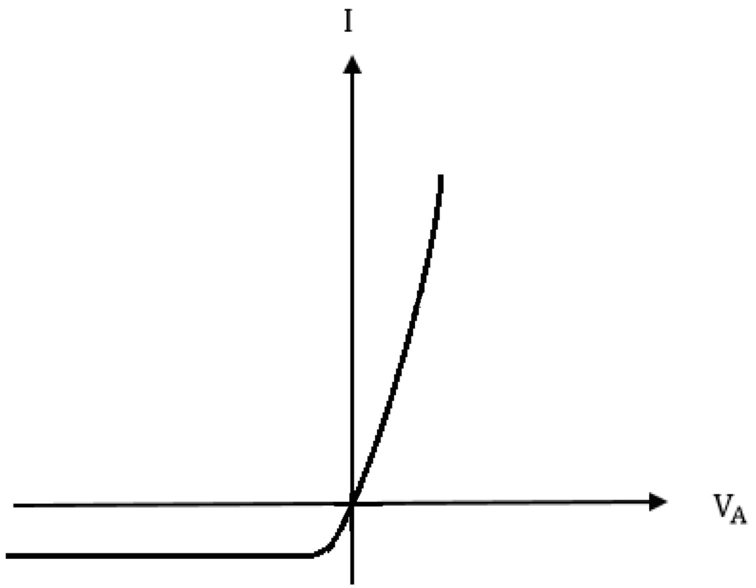
Schottky Barrier Contact Under bias V

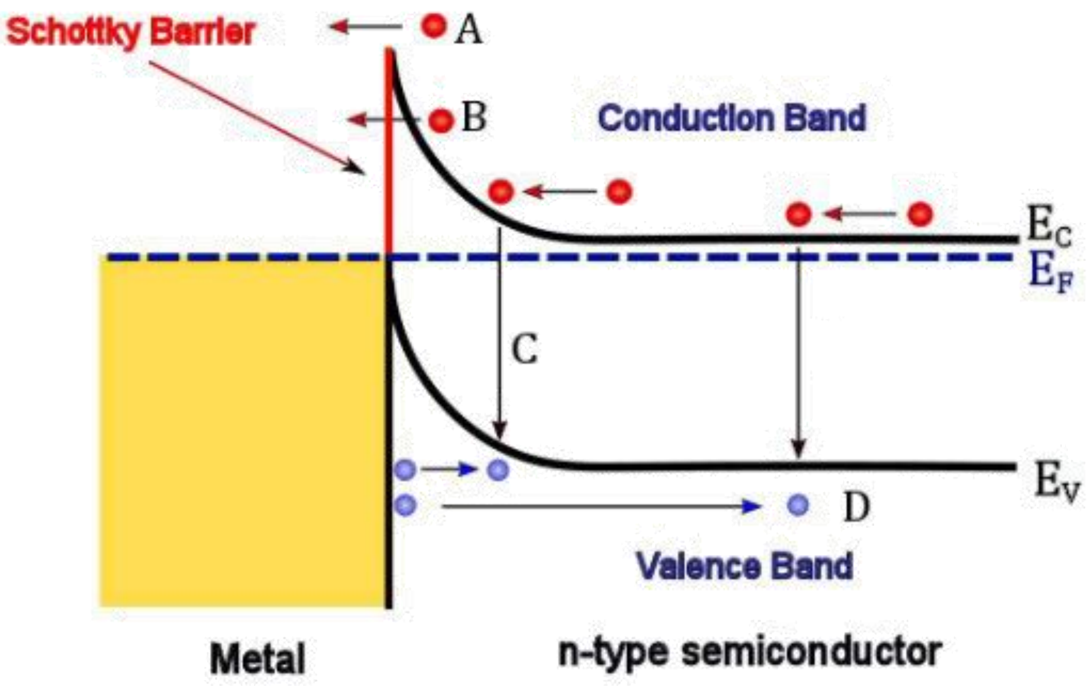
Reverse Biasing



Schottky Barrier Contact Under bias V





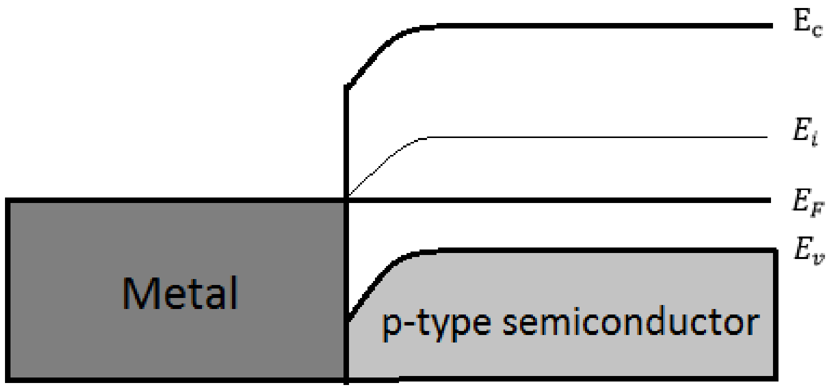


<u>Current transport mechanism across Schottky diode</u>

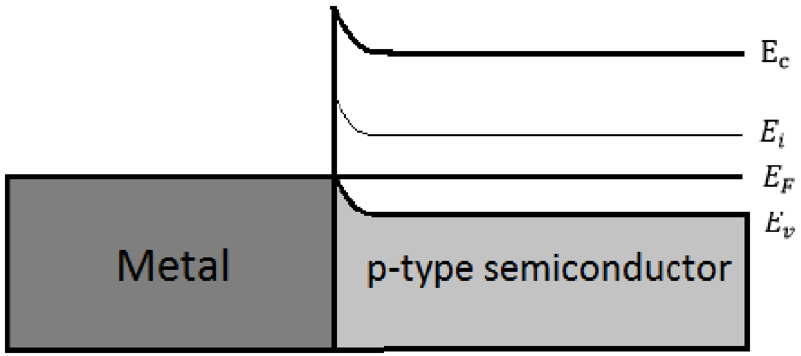
(A) thermionic emission,(B) tunnelling, (C) carrier recombination in the depletion region and (D) hole injection from metal

Equilibrium energy band diagram for a metal to an p-type semiconductor

For metal and p-type semiconductor, the equilibrium energy band diagrams are shown below:



$\Phi_M < \Phi_S$



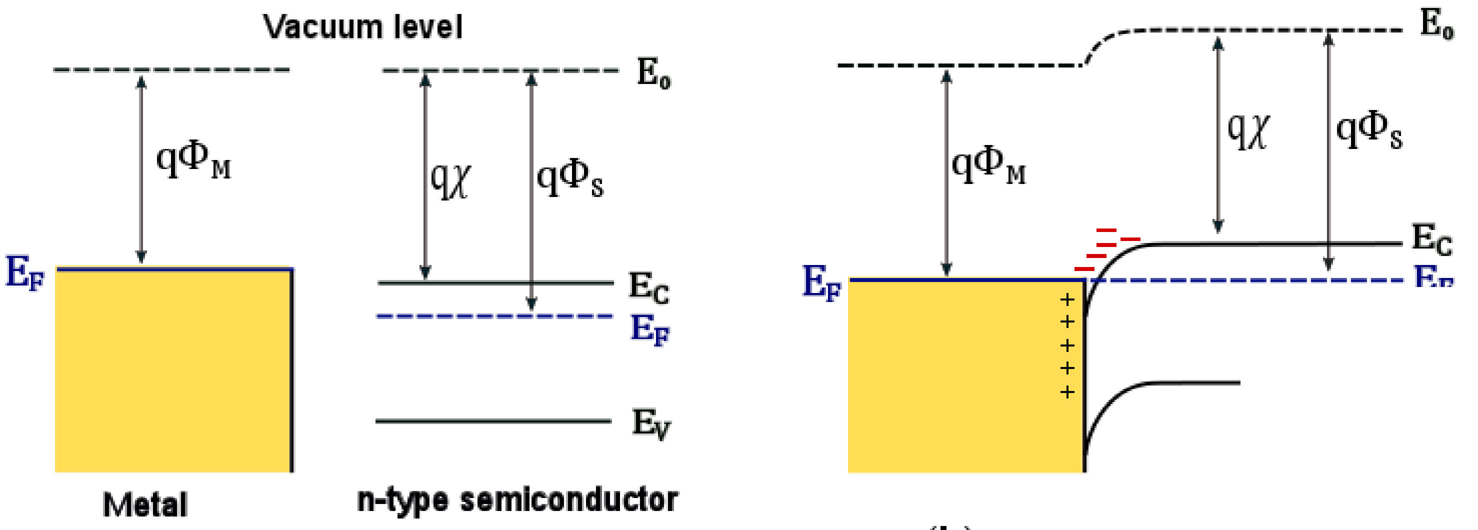
$\Phi_M > \Phi_S$

and for p type Semiconductor

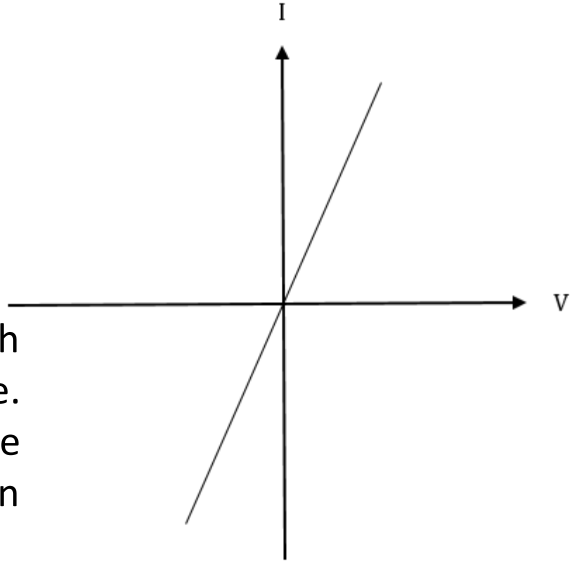
$$\Phi_B = Eg + \chi - \Phi_M$$

Ohmic Contact

Not all MS contact can perform as the rectifying Schottky diode, since there is no potential barrier formed. Under this situation, when the current can be conducted in both directions of the MS contact, the contact is defined as the Ohmic contact.



(a) (b)



An ideal Ohmic contact is a low resistance, and non-rectifying junction with $\Phi_M < \Phi_S$; no potential exists between the metal-semiconductor interface. The MS contact with electrons will transfer from the metal to the semiconductor due to their low energy, which will cause the Fermi level in semiconductor move up until the equilibrium state is established

Electrical nature of MS contacts

	n-type semiconductor	p-type semiconductor
$\Phi_M > \Phi_S$	Rectifying	Ohmic
$\Phi_M < \Phi_S$	Ohmic	Rectifying

Metal	Work function ϕ_M [eV]	*n*-type Si	*p*-type Si
Pt	6.30	Schottky contact $\phi_B = 6.30 - 4.05 = 2.25$ eV	Ohmic contact ϕ_M (= 6.30) > $\chi + E_g$ (= 4.05 + 1.11)
Ni	4.01	Ohmic contact ϕ_M (= 4.01) < χ (=4.05)	Schottky contact $\phi_B = 4.05 + 1.11 - 4.01 = 1.15$ eV

Nonlinear Ohmic contact/ tunnel contact

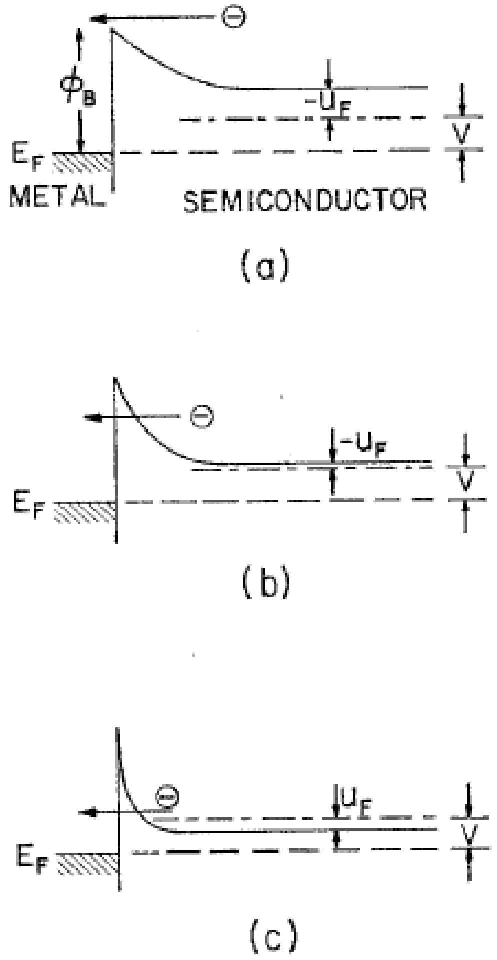


FIG. 1 Band diagrams of metal n-type semiconductor contacts under forward bias voltage: (a) semiconductor lightly doped, $kT/E_{00} \gg 1$, (b) semiconductor heavily doped, $kT/E_{00} \sim 1$, (c) semiconductor very heavily doped (degenerate), $kT/E_{00} \ll 1$.